

JEE MAIN 2027 EDITION · SESSION 1: 22-30 JAN 2027 (TENTATIVE)

JEE MAIN

99+

THE COMPLETE PERSONALIZED PREPARATION
SYSTEM

*A Personalized Strategy, Formula, PYQ, Practice, Revision &
Mock-Test Handbook*

● Physics

● Chemistry

● Mathematics

● Strategy · PYQs · Mocks

PREPARED FOR

Gayathri Devu

QUICK REVISION BOOK

Formula handbook · Speed & shortcut manual ·
Last-minute sheets

Contents

Page numbers marked **MB** refer to the Master Book, where the full chapter is.

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PART V



Master Formula Handbook

Every formula, constant, identity, reaction and trend, organised for 1-day, 3-day, 7-day and final 24-hour revision.

• How to Use the Formula Handbook	4
• Physics Formula Sheets	5
• Chemistry Formula & Fact Sheets	12
• Mathematics Formula Sheets	17

• Constants, Units & Frequently Forgotten Formulas	20
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PART V · MASTER FORMULA HANDBOOK

How to Use the Formula Handbook

This part compresses the formula content of all 54 modules into sheets you can revise in minutes. Nothing here is new: every line comes from the chapter modules, where the derivations, conditions and examples are. Use it for revision, not for first learning.

Physics sheets · page [5](#)

All 20 units, with sign conventions and the conditions each formula needs.

Chemistry sheets · page [12](#)

Physical formulas, inorganic trends and exceptions, organic reagents and named reactions.

Maths sheets · page [17](#)

All 14 units: identities, standard results, integrals and conic parameters.

Constants & frequently forgotten formulas · page [20](#)

[20](#)

Physical constants (CODATA values), unit conversions, and the formulas students forget most often.

Formula revision plans · page [23](#)

[23](#)

1-Day, 3-Day and 7-Day cycles and the Final 24-Hour plan, with checkboxes.

The three-pass method RECOMMENDATION

1 Pass 1 · Recall

Cover the right column of a sheet and write each formula from memory on a rough sheet. Retrieval practice is one of the best-supported learning techniques [LS-DUN], [LS-RK].

2 Pass 2 · Check & mark

Uncover and compare. Put a dot next to every formula you got wrong or hesitated on.

3 Pass 3 · Repeat the dots

Re-test only the dotted formulas the next day, then after 3 and 7 days (spaced revision, page [MB 449](#)). A formula with three dots goes into the Frequently Forgotten list.

☆ IMPORTANT Conditions are part of the formula

Most formula errors in JEE come from using a correct formula outside its conditions: $v = u + at$ only for constant acceleration; S_∞ only for $|r| < 1$; $\Delta G^\circ = -RT \ln K$ uses the standard-state K . The sheets list the condition next to each formula where it matters. Revise the condition along with the formula.

💡 TIP Printing

The sheets are designed for A4 print at 100% scale. For a pocket version, print two pages per sheet (A5 each). Text stays readable down to about 70%.

PART V · MASTER FORMULA HANDBOOK

Physics Formula Sheets

Each sheet links to its full module. The symbols follow NCERT usage.

Σ FORMULA P01 · Units & Measurements · page MB 36

Result	Formula / rule
Relative error of $Q = \frac{a^p b^q}{c^r}$	$\frac{\Delta Q}{Q} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$ (errors always add)
Sum or difference $Q = a \pm b$	$\Delta Q = \Delta a + \Delta b$ (absolute errors add)
Least count, vernier	$LC = 1 \text{ MSD} - 1 \text{ VSD} = \frac{1 \text{ MSD}}{n}$
Least count, screw gauge	$LC = \frac{\text{pitch}}{\text{number of circular divisions}}$
Useful dimensions	$[G] = \text{M}^{-1} \text{L}^3 \text{T}^{-2}$ · $[h] = \text{ML}^2 \text{T}^{-1}$ · $[\epsilon_0] = \text{M}^{-1} \text{L}^{-3} \text{T}^4 \text{A}^2$ · $[\mu_0] = \text{MLT}^{-2} \text{A}^{-2}$
Significant figures	$\times/\div$: keep the fewest significant figures · $+/ -$: keep the fewest decimal places

Σ FORMULA P02 · Kinematics · page MB 43

Result	Formula	Condition
Equations of motion	$v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$	constant a
Distance in the n th second	$s_n = u + \frac{a}{2}(2n - 1)$	constant a
Projectile	$T = \frac{2u \sin \theta}{g}$ · $H = \frac{u^2 \sin^2 \theta}{2g}$ · $R = \frac{u^2 \sin 2\theta}{g}$	level ground
Trajectory	$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R}\right)$	—
Relative velocity; river	$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ · shortest time $t = d/v_{br}$ · zero drift if $\sin \theta = v_r/v_{br}$	—
Circular motion	$a_c = \frac{v^2}{r} = \omega^2 r$ · $a_t = \frac{dv}{dt}$ · $a = \sqrt{a_c^2 + a_t^2}$	—

Σ FORMULA P03 · Laws of Motion · page MB 50

Result	Formula
Newton's second law	$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$ (constant mass) · impulse $= \Delta p = \int F dt$
Friction	$f_s \leq \mu_s N$ · $f_k = \mu_k N$ · angle of repose $\tan \theta = \mu_s$
Incline (rough)	block slides if $\tan \theta > \mu_s$ · $a = g(\sin \theta - \mu_k \cos \theta)$
Atwood machine	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$ · $T = \frac{2m_1 m_2 g}{m_1 + m_2}$
Banking	$\tan \theta = \frac{v^2}{rg}$ (no friction) · $v_{max} = \sqrt{rg \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}$
Flat curve	$v_{max} = \sqrt{\mu rg}$
Vertical circle (string)	$v_{bottom} \geq \sqrt{5gl}$ · $v_{top} \geq \sqrt{gl}$ · $T_{bottom} - T_{top} = 6mg$

Σ FORMULA P04 · Work, Energy & Power · page MB 57

Result	Formula
Work	$W = \int \vec{F} \cdot d\vec{s}$ · work-energy theorem $W_{net} = \Delta K$
Spring	$U = \frac{1}{2}kx^2$ · springs in series $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ · parallel $k = k_1 + k_2$
Conservative force	$F = -\frac{dU}{dx}$ · stable equilibrium where $\frac{d^2U}{dx^2} > 0$
Power	$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$
Collisions (1D)	$e = \frac{\text{separation speed}}{\text{approach speed}}$ · elastic, equal masses: velocities exchange
Elastic collision, m_2 at rest	$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1$ · $v_2 = \frac{2m_1}{m_1 + m_2}u_1$
KE-momentum	$K = \frac{p^2}{2m}$

Σ FORMULA P05 · Rotational Motion & Centre of Mass · page MB 63

Result	Formula
Centre of mass	$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$ · semicircular ring $\frac{2R}{\pi}$ · semicircular disc $\frac{4R}{3\pi}$ · hemisphere (solid) $\frac{3R}{8}$ · cone $\frac{h}{4}$ from base
Moments of inertia	ring MR^2 · disc $\frac{1}{2}MR^2$ · solid sphere $\frac{2}{5}MR^2$ · hollow sphere $\frac{2}{3}MR^2$ · rod (centre) $\frac{1}{12}ML^2$ · rod (end) $\frac{1}{3}ML^2$
Axis theorems	parallel $I = I_{cm} + Md^2$ · perpendicular $I_z = I_x + I_y$ (planar bodies)
Rotation	$\tau = I\alpha$ · $L = I\omega$ · $K = \frac{1}{2}I\omega^2$ · $\tau = \frac{dL}{dt}$ (L conserved if $\tau_{ext} = 0$)
Rolling without slipping	$v = \omega R$ · $K = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2}\right)$ · on an incline $a = \frac{g \sin \theta}{1 + k^2/R^2}$
Radius of gyration	$I = Mk^2$

Σ FORMULA P06 · Gravitation · page MB 70

Result	Formula
Field & potential	$g = \frac{GM}{R^2}$ · $V = -\frac{GM}{r}$ · $U = -\frac{GMm}{r}$
Variation of g	height $g_h = \frac{g}{(1 + h/R)^2} \approx g \left(1 - \frac{2h}{R}\right)$ · depth $g_d = g \left(1 - \frac{d}{R}\right)$
Orbits	$v_o = \sqrt{\frac{GM}{r}}$ · $T^2 = \frac{4\pi^2}{GM}r^3$ · $E = -\frac{GMm}{2r}$ · $K = -E$, $U = 2E$
Escape speed	$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} = \sqrt{2}v_o$ (surface orbit) · Earth: ≈ 11.2 km/s
Kepler II	areal velocity $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$
Inside a uniform sphere	$g = \frac{GMr}{R^3}$ · shell: $g = 0$ inside

Σ FORMULA P07 · Properties of Solids & Liquids · page MB 76

Result	Formula
Elasticity	$Y = \frac{FL}{A\Delta L}$ · $B = -\frac{\Delta P}{\Delta V/V}$ · energy density = $\frac{1}{2}$ stress $\times$ strain
Fluids	$P = P_0 + \rho gh$ · continuity $A_1v_1 = A_2v_2$ · Bernoulli $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ · Torricelli $v = \sqrt{2gh}$
Viscosity	$F = \eta A \frac{dv}{dx}$ · Stokes $F = 6\pi\eta rv$ · terminal $v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$
Surface tension	excess pressure: drop $\frac{2T}{r}$, bubble $\frac{4T}{r}$ · capillary rise $h = \frac{2T \cos \theta}{r\rho g}$
Thermal expansion	$\Delta L = L\alpha\Delta T$ · $\beta = 2\alpha$ · $\gamma = 3\alpha$ · thermal stress = $Y\alpha\Delta T$
Heat	$Q = mc\Delta T$ · $Q = mL$ · conduction $\frac{dQ}{dt} = \frac{kA\Delta T}{L}$ · $R_{th} = \frac{L}{kA}$
Radiation & cooling	Stefan $P = \sigma eAT^4$ · Wien $\lambda_m T = b$ · Newton $\frac{dT}{dt} = -k(T - T_0)$

Σ FORMULA P08 · Thermodynamics · page MB 84

Process	Work by gas W	Key relation
First law	—	$\Delta Q = \Delta U + W$ · $\Delta U = nC_V\Delta T$ always (ideal gas)
Isothermal	$nRT \ln \frac{V_2}{V_1}$	$PV = \text{const}$ · $\Delta U = 0$
Adiabatic	$\frac{P_1V_1 - P_2V_2}{\gamma - 1}$	$PV^\gamma = \text{const}$ · $TV^{\gamma-1} = \text{const}$ · $Q = 0$
Isobaric	$P\Delta V = nR\Delta T$	$Q = nC_P\Delta T$
Isochoric	0	$Q = nC_V\Delta T$
Cyclic	area enclosed	$\Delta U = 0$
Carnot efficiency $\eta = 1 - \frac{T_2}{T_1}$ · refrigerator COP = $\frac{T_2}{T_1 - T_2}$ · $C_P - C_V = R$ · slope: adiabatic = $\gamma \times$ isothermal.		

Σ FORMULA P09 · Kinetic Theory of Gases · page MB 90

Result	Formula
Pressure	$P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3}\frac{mN}{V}v_{rms}^2$
Speeds	$v_{rms} = \sqrt{\frac{3RT}{M}}$ · $v_{avg} = \sqrt{\frac{8RT}{\pi M}}$ · $v_{mp} = \sqrt{\frac{2RT}{M}}$ ($v_{rms} > v_{avg} > v_{mp}$)
Energy	average KE per molecule = $\frac{3}{2}k_B T$ · equipartition $\frac{1}{2}k_B T$ per degree of freedom
Specific heats	$C_V = \frac{f}{2}R$ · $\gamma = 1 + \frac{2}{f}$ · monatomic $f = 3$ ($\gamma = \frac{5}{3}$) · diatomic $f = 5$ ($\gamma = \frac{7}{5}$)
Mixture	$C_{V,mix} = \frac{n_1C_{V1} + n_2C_{V2}}{n_1 + n_2}$
Mean free path	$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

Σ FORMULA P10 · Oscillations & Waves · page MB 95

Result	Formula
SHM	$x = A \sin(\omega t + \phi) \cdot v = \omega \sqrt{A^2 - x^2} \cdot a = -\omega^2 x \cdot E = \frac{1}{2} m \omega^2 A^2$
Periods	spring $T = 2\pi \sqrt{\frac{m}{k}}$ · pendulum $T = 2\pi \sqrt{\frac{l}{g}}$ · in a lift $g_{eff} = g \pm a$
Wave speed	string $v = \sqrt{\frac{T}{\mu}}$ · sound $v = \sqrt{\frac{\gamma P}{\rho}}$ · $v = f \lambda$
Strings and pipes	string / open pipe $f_n = \frac{nv}{2L}$ (all harmonics) · closed pipe $f_n = \frac{nv}{4L}$, n odd
Beats & Doppler	$f_{beat} = f_1 - f_2 \cdot f' = f \frac{v \pm v_o}{v \mp v_s}$ (upper signs: approaching)
Travelling wave	$y = A \sin(kx - \omega t) \cdot v = \frac{\omega}{k} \cdot \text{intensity} \propto A^2 \omega^2$

Σ FORMULA P11 · Electrostatics & Capacitors · page MB 101

Result	Formula
Coulomb, field, potential	$F = \frac{kq_1 q_2}{r^2} \cdot E = \frac{kq}{r^2} \cdot V = \frac{kq}{r} \cdot E = -\frac{dV}{dr} \cdot k = \frac{1}{4\pi\epsilon_0}$
Dipole	axial $E = \frac{2kp}{r^3}$ · equatorial $\frac{kp}{r^3}$ · $\tau = \vec{p} \times \vec{E} \cdot U = -\vec{p} \cdot \vec{E}$
Gauss's law	$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$ · line $\frac{\lambda}{2\pi\epsilon_0 r}$ · sheet $\frac{\sigma}{2\epsilon_0}$ · conductor surface $\frac{\sigma}{\epsilon_0}$
Capacitors	$C = \frac{\epsilon_0 A}{d}$ · with dielectric K · slab: $C = \frac{\epsilon_0 A}{d - t + t/K}$ · sphere $4\pi\epsilon_0 R$
Energy	$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$ · energy density $\frac{1}{2} \epsilon_0 E^2$
Combinations	series $\frac{1}{C} = \sum \frac{1}{C_i}$ · parallel $C = \sum C_i$ · sharing: common $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

Σ FORMULA P12 · Current Electricity · page MB 108

Result	Formula
Current & drift	$I = n e A v_d \cdot v_d = \frac{e E \tau}{m} \cdot \rho = \frac{m}{n e^2 \tau} \cdot R = \frac{\rho l}{A}$
Stretched wire	length $\times n$ at constant volume: $R \rightarrow n^2 R$
Temperature	$R_T = R_0(1 + \alpha \Delta T)$
Cell	$V = \epsilon - I r \cdot P_{max}$ to load when $R = r$ · cells in series $\epsilon_{eq} = \sum \epsilon_i$, $r_{eq} = \sum r_i$
Cells in parallel	$\epsilon_{eq} = \frac{\sum \epsilon_i / r_i}{\sum 1/r_i}$
Bridges & meters	Wheatstone balance $\frac{P}{Q} = \frac{R}{S}$ · meter bridge $\frac{R}{S} = \frac{l}{100 - l}$ · potentiometer $\frac{\epsilon_1}{\epsilon_2} = \frac{l_1}{l_2} \cdot r = R \left(\frac{l_1}{l_2} - 1 \right)$
Power	$P = VI = I^2 R = \frac{V^2}{R}$

Σ FORMULA P13 · Magnetic Effects & Magnetism · page MB 114

Result	Formula
Fields	long wire $\frac{\mu_0 I}{2\pi r}$ · loop centre $\frac{\mu_0 I}{2R}$ · axis $\frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$ · solenoid $\mu_0 n I$ · arc $\frac{\mu_0 I \theta}{4\pi R}$
Forces	$\vec{F} = q\vec{v} \times \vec{B}$ · $\vec{F} = I\vec{L} \times \vec{B}$ · parallel wires $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$
Charge in B	$r = \frac{mv}{qB}$ · $T = \frac{2\pi m}{qB}$ (independent of v) · pitch = $v_{\parallel} T$
Loop	$\vec{m} = NI\vec{A}$ · $\tau = \vec{m} \times \vec{B}$ · $U = -\vec{m} \cdot \vec{B}$
Galvanometer conversions	ammeter: shunt $S = \frac{I_g G}{I - I_g}$ · voltmeter: series $R = \frac{V}{I_g} - G$
Magnetic materials	$B = \mu_0(H + M)$ · $\chi = \frac{M}{H}$ · $\mu_r = 1 + \chi$ · Curie $\chi \propto \frac{1}{T}$ (paramagnets)

Σ FORMULA P14 · EMI & Alternating Current · page MB 121

Result	Formula
Faraday & motional emf	$\varepsilon = -\frac{d\Phi}{dt}$ · rod $\varepsilon = Blv$ · rotating rod $\frac{1}{2}B\omega l^2$ · coil $NBA\omega \sin \omega t$
Inductance	$L = \mu_0 n^2 Al$ · $U = \frac{1}{2}LI^2$ · $M = \mu_0 n_1 n_2 Al$
Transients	LR: $I = I_0(1 - e^{-t/\tau})$, $\tau = \frac{L}{R}$ · RC: $q = CV(1 - e^{-t/RC})$
AC	$V_{rms} = \frac{V_0}{\sqrt{2}}$ · $X_L = \omega L$ · $X_C = \frac{1}{\omega C}$ · $Z = \sqrt{R^2 + (X_L - X_C)^2}$ · $\tan \phi = \frac{X_L - X_C}{R}$
Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}$ · $Q = \frac{\omega_0 L}{R}$ · bandwidth = $\frac{R}{L}$
Power	$P = V_{rms} I_{rms} \cos \phi$ · power factor $\cos \phi = \frac{R}{Z}$
Transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ (ideal)

Σ FORMULA P15 · Electromagnetic Waves · page MB 128

Result	Formula
Speed & ratio	$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \cdot \frac{E_0}{B_0} = c$ · in a medium $v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$
Displacement current	$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$
Energy	$u_{avg} = \frac{1}{2}\varepsilon_0 E_0^2$ · $I = u_{avg} c$ · radiation pressure $\frac{I}{c}$ (absorbed), $\frac{2I}{c}$ (reflected)
Spectrum (increasing λ)	γ · X-rays · UV · visible · IR · microwaves · radio

Σ FORMULA P16 · Optics: Ray & Wave · page MB 132

Result	Formula
Mirrors & lenses	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (mirror) · $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ (lens) · $m = -\frac{v}{u}$ (mirror), $\frac{v}{u}$ (lens)
Lens maker	$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ · combination $P = \sum P_i$
Refraction	Snell $n_1 \sin i = n_2 \sin r$ · apparent depth $= \frac{\text{real}}{n}$ · $\sin C = \frac{1}{n}$ · curved surface $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$
Prism	$A = r_1 + r_2$ · $\delta = i + e$ · $A \cdot n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$ · thin prism $\delta = (n-1)A$
Instruments	microscope $m = \frac{L}{f_o} \frac{D}{f_e}$ · telescope $m = \frac{f_o}{f_e}$ (normal adjustment)
YDSE	$\beta = \frac{\lambda D}{d}$ · $I = 4I_0 \cos^2 \frac{\phi}{2}$ · path difference $\Delta x = \frac{yd}{D}$ · sheet shift $\frac{(\mu-1)tD}{d}$
Diffraction & polarisation	single slit minima $a \sin \theta = n\lambda$ · central width $\frac{2\lambda D}{a}$ · Malus $I = I_0 \cos^2 \theta$ · Brewster $\tan i_B = n$

Σ FORMULA P17 · Dual Nature · page MB 139

Result	Formula
Photon	$E = h\nu = \frac{hc}{\lambda}$ · $E(\text{eV}) \approx \frac{1240}{\lambda(\text{nm})}$ · $p = \frac{h}{\lambda}$
Photoelectric	$K_{max} = h\nu - \phi = eV_0$ · threshold $\nu_0 = \frac{\phi}{h}$ · the stopping potential depends on frequency, not intensity
de Broglie	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$ · electron: $\lambda(\text{\AA}) \approx \frac{12.27}{\sqrt{V}}$

Σ FORMULA P18 · Atoms & Nuclei · page MB 144

Result	Formula
Bohr (hydrogen-like)	$r_n = 0.529 \frac{n^2}{Z} \text{\AA}$ · $v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$ · $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$
Spectrum	$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ · lines from level n : $\frac{n(n-1)}{2}$
Nucleus	$R = R_0 A^{1/3}$ (density independent of A) · $BE = \Delta m c^2$ · $1 \text{ u} = 931.5 \text{ MeV}/c^2$
Decay	$N = N_0 e^{-\lambda t}$ · $T_{1/2} = \frac{\ln 2}{\lambda}$ · mean life $\tau = \frac{1}{\lambda}$ · activity $A = \lambda N$

Σ FORMULA P19 · Electronic Devices · page MB 150

Result	Formula / fact
Semiconductors	$n_e n_h = n_i^2$ · n-type: pentavalent donor · p-type: trivalent acceptor
Diode	forward bias: barrier falls · reverse: tiny current until breakdown · Zener: regulator in reverse bias
Rectifiers	half-wave output frequency f · full-wave $2f$
Logic gates	NAND and NOR are universal · $\overline{A \cdot B} = \bar{A} + \bar{B}$ · $\overline{A + B} = \bar{A} \cdot \bar{B}$

Σ FORMULA P20 · Experimental Skills · page MB 155

Experiment	Key relation
Vernier / screw gauge	reading = MSR + (coincidence $\times$ LC) – zero error (with sign)
Simple pendulum	slope of T^2 vs $l = \frac{4\pi^2}{g}$
Young's modulus (Searle)	$Y = \frac{MgL}{\pi r^2 \Delta l}$
Surface tension (capillary)	$T = \frac{r h \rho g}{2 \cos \theta}$
Resonance tube	$\lambda = 2(l_2 - l_1)$ · end correction $e = \frac{l_2 - 3l_1}{2}$
Metre bridge	$X = R \frac{l}{100 - l}$
Focal length (u–v)	$f = \frac{uv}{u + v}$ (magnitudes, real image)

PART V · MASTER FORMULA HANDBOOK

Chemistry Formula & Fact Sheets

Physical Chemistry

Σ FORMULA C01-C02 · Mole concept & atomic structure · pages MB 165, MB 170

Result	Formula
Moles	$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V_{\text{gas, STP}}}{22.7 \text{ L}}$ (IUPAC STP, 1 bar) · 22.4 L at 1 atm and 273 K
Concentration	$M = \frac{n_{\text{solute}}}{V(\text{L})} \cdot m = \frac{n_{\text{solute}}}{\text{kg solvent}} \cdot M = \frac{10 x d}{M_{\text{solute}}}$ ($x = \% \text{ w/w}$, d in g/mL)
Equivalents	n -factor method: equivalents = $n \times n\text{-factor} \cdot N = M \times n\text{-factor}$
Photon & Bohr	$E = h\nu \cdot E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \cdot r_n = 0.529 \frac{n^2}{Z} \text{ Å}$
Uncertainty & de Broglie	$\Delta x \Delta p \geq \frac{h}{4\pi} \cdot \lambda = \frac{h}{p}$
Nodes	radial $n - l - 1$ · angular l · total $n - 1$
Orbital angular momentum	$\sqrt{l(l+1)} \frac{h}{2\pi}$ · spin-only moment $\mu = \sqrt{n(n+2)} \text{ BM}$

Σ FORMULA C03 · Chemical bonding · page MB 175

Result	Formula / rule
Hybridisation (central atom)	steric number = bond pairs + lone pairs: $2 sp \cdot 3 sp^2 \cdot 4 sp^3 \cdot 5 sp^3 d \cdot 6 sp^3 d^2$
Bond order (MOT)	$\frac{1}{2}(N_b - N_a)$ · paramagnetic if unpaired MO electrons: $\text{O}_2, \text{B}_2, \text{O}_2^-$
Dipole moment	$\mu = q \times d$ · zero for symmetrical shapes ($\text{CO}_2, \text{BF}_3, \text{CH}_4, \text{SF}_6, \text{XeF}_4$)
Formal charge	$V - L - \frac{1}{2}B$
Bond angle order	$\text{CH}_4 (109.5^\circ) > \text{NH}_3 (107^\circ) > \text{H}_2\text{O} (104.5^\circ)$ — lone pairs compress angles

Σ FORMULA C04 · Thermodynamics · page MB 181

Result	Formula
First law	$\Delta U = q + w$ (IUPAC sign: w done on the system) · $w = -P_{\text{ext}}\Delta V$
Enthalpy	$\Delta H = \Delta U + \Delta n_g RT$
Reversible isothermal	$w = -2.303 nRT \log \frac{V_2}{V_1}$
Hess & bond enthalpy	$\Delta_r H = \sum \Delta_f H(\text{products}) - \sum \Delta_f H(\text{reactants}) = \sum BE(\text{broken}) - \sum BE(\text{formed})$
Entropy	$\Delta S = \frac{q_{\text{rev}}}{T}$ · phase change $\Delta S = \frac{\Delta H}{T}$
Gibbs	$\Delta G = \Delta H - T\Delta S \cdot \Delta G^\circ = -RT \ln K = -2.303 RT \log K \cdot \Delta G^\circ = -nFE^\circ$
Spontaneity	$\Delta H < 0, \Delta S > 0$: always · $\Delta H > 0, \Delta S > 0$: above $T = \frac{\Delta H}{\Delta S}$

Σ FORMULA C05 · Solutions · page MB 186

Property	Formula
Raoult's law	$p_A = x_A p_A^\circ$ · relative lowering $\frac{p^\circ - p}{p^\circ} = x_{\text{solute}}$
Elevation / depression	$\Delta T_b = i K_b m$ · $\Delta T_f = i K_f m$
Osmotic pressure	$\pi = i C R T$
van 't Hoff factor	association $i = 1 - \alpha \left(1 - \frac{1}{n}\right)$ · dissociation $i = 1 + (n - 1)\alpha$
Henry's law	$p = K_H x$ (higher $K_H \rightarrow$ lower solubility)
Deviations	positive (A-B weaker than A-A, B-B): $\Delta H_{\text{mix}} > 0$, minimum-boiling azeotrope · negative: $\Delta H_{\text{mix}} < 0$, maximum-boiling azeotrope

Σ FORMULA C06 · Equilibrium · page MB 191

Result	Formula
K_p and K_c	$K_p = K_c (RT)^{\Delta n_g}$
Reaction quotient	$Q < K$: forward · $Q > K$: backward
Weak acid	$[\text{H}^+] = \sqrt{K_a C}$ · $\alpha = \sqrt{K_a / C}$ (Ostwald, small α)
pH	$\text{pH} + \text{pOH} = 14$ (25 °C) · $K_w = 10^{-14}$
Buffers	$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$ · $\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$
Salt hydrolysis	WA-SB: $\text{pH} = 7 + \frac{1}{2}(\text{p}K_a + \log C)$ · SA-WB: $7 - \frac{1}{2}(\text{p}K_b + \log C)$ · WA-WB: $7 + \frac{1}{2}(\text{p}K_a - \text{p}K_b)$
Solubility	AB: $K_{sp} = s^2$ · AB ₂ : $4s^3$ · A ₂ B ₃ : $108s^5$ · precipitation if $Q_{sp} > K_{sp}$

Σ FORMULA C07 · Electrochemistry · page MB 197

Result	Formula
Cell emf	$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$ (reduction potentials)
Nernst	$E = E^\circ - \frac{0.0591}{n} \log Q$ (298 K) · at equilibrium $\log K = \frac{nE^\circ}{0.0591}$
Conductance	$\kappa = \frac{G^*}{R}$ ($G^* = l/A$) · $\Lambda_m = \frac{1000 \kappa}{M}$ (S cm ² mol ⁻¹)
Kohlrausch	$\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$ (strong) · $\Lambda_m^\circ = \sum \nu_i \lambda_i^\circ$ · $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$
Faraday	$m = \frac{MIt}{nF}$ · $F = 96485$ C/mol

Σ FORMULA C08 · Chemical kinetics · page MB 203

Order	Integrated law	Half-life	Units of k
0	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$	mol L ⁻¹ s ⁻¹
1	$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$	$\frac{0.693}{k}$	s ⁻¹
n	—	$\propto [A]_0^{1-n}$	(mol L ⁻¹) ¹⁻ⁿ s ⁻¹

Arrhenius $k = Ae^{-E_a/RT}$ · $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ · first order: 99.9% completion $\approx 10 t_{1/2}$.

Inorganic Chemistry: trends & exceptions

REFERENCE C09 · Periodicity · page [MB 208](#)

Property	Trend	Exceptions to remember
Atomic radius	decreases across a period, increases down a group	Ga $\approx$ Al (d-contraction) · 4d $\approx$ 5d (lanthanoid contraction)
Ionisation enthalpy	increases across a period	Be > B, N > O (and Mg > Al, P > S)
Electron gain enthalpy	more negative across a period	Cl > F (most negative is Cl) · noble gases and N: positive or near zero
Electronegativity	increases across a period	F highest (4.0, Pauling)
Isoelectronic ions	size decreases as Z increases	$\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

REFERENCE C10–C12 · p-, d-, f-block & coordination · pages [MB 213](#), [MB 219](#), [MB 225](#)

Topic	Key facts
Inert-pair effect	stability of the lower state increases down groups 13–15: Tl^+ , Pb^{2+} , Bi^{3+}
Oxoacid strength	$\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$ · hydrogen-halide acidity $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
Hydrides	thermal stability $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3$ · boiling point $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
d-block	magnetic moment $\sqrt{n(n+2)}$ BM · Mn shows +7 · KMnO_4 : acidic $\rightarrow \text{Mn}^{2+}$ ($n = 5$), neutral/weakly basic $\rightarrow \text{MnO}_2$ ($n = 3$)
$\text{K}_2\text{Cr}_2\text{O}_7$	orange; $\text{Cr}_2\text{O}_7^{2-} \rightleftharpoons \text{CrO}_4^{2-}$ (yellow) with pH · acidic $\rightarrow \text{Cr}^{3+}$ ($n = 6$)
Lanthanoids	common state +3 · Ce +4 (oxidant) · Eu, Yb +2 · lanthanoid contraction
Coordination: CFSE (octahedral)	$(-0.4n_{t_{2g}} + 0.6n_{e_g})\Delta_o \cdot \Delta_t = \frac{4}{9}\Delta_o$
Spectrochemical series	$\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^- < \text{OH}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
Isomer counts	$[\text{MA}_4\text{B}_2]$: 2 geometric · $[\text{MA}_3\text{B}_3]$: fac/mer · $[\text{M}(\text{AA})_2\text{B}_2]$: cis is optically active (3 stereoisomers)

Organic Chemistry

REFERENCE C14 · GOC orders · page [MB 237](#)

Quantity	Order
Carbocation stability	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$ · benzylic/allylic stabilised by resonance
Carbanion stability	$\text{CH}_3^- > 1^\circ > 2^\circ > 3^\circ$ (opposite of carbocations)
Free radicals	$3^\circ > 2^\circ > 1^\circ$ · benzyl $\approx$ allyl $> 3^\circ$
Acidity	$\text{RCOOH} > \text{phenol} > \text{H}_2\text{O} > \text{ROH} > \text{alkyne} > \text{alkene} > \text{alkane}$ · EWG increases acidity
Basicity (aqueous)	$(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{aniline}$
Effects	$-I$: $\text{NO}_2 > \text{CN} > \text{COOH} > \text{F} > \text{Cl}$ · $+M$: $-\text{NH}_2, -\text{OH}, -\text{OR}$

REFERENCE Reagent table (C15–C18) · pages MB 242–MB 264

Reagent	Does
H ₂ /Lindlar (Pd–BaSO ₄ , quinoline)	alkyne → <i>cis</i> -alkene
Na/liquid NH ₃	alkyne → <i>trans</i> -alkene
HBr + peroxide	anti-Markovnikov addition (HBr only)
Alkaline KMnO ₄ (cold, dilute)	Baeyer's test: alkene → <i>syn</i> -diol
O ₃ , then Zn/H ₂ O	ozonolysis → aldehydes/ketones
LiAlH ₄	reduces –COOH, esters, aldehydes, ketones, amides and nitriles
NaBH ₄	reduces aldehydes and ketones only
PCC	1° alcohol → aldehyde (stops there)
K ₂ Cr ₂ O ₇ /H ⁺	1° alcohol → acid · 2° → ketone
DIBAL-H	ester / nitrile → aldehyde
SOCl ₂ (pyridine)	ROH → RCl (by-products SO ₂ , HCl are gases)
alcoholic KOH	dehydrohalogenation (elimination)
aqueous KOH	substitution (RX → ROH)
Br ₂ /NaOH	Hofmann bromamide: amide → amine with one C fewer
NaNO ₂ /HCl, 0–5 °C	diazotisation of aromatic 1° amines
Sn/HCl or Fe/HCl	ArNO ₂ → ArNH ₂

REFERENCE Named reactions (C15–C18)

Reaction	Change
Wurtz	2RX + 2Na → R–R (dry ether)
Friedel–Crafts	ArH + RCl/RCOCl + anhydrous AlCl ₃ → alkyl/acyl arene
Kolbe (electrolytic)	RCOONa → R–R
Kolbe (phenol)	phenoxide + CO ₂ → salicylic acid
Reimer–Tiemann	phenol + CHCl ₃ /NaOH → salicylaldehyde
Williamson	RONa + R'X → ROR' (R'X must be 1°)
Aldol	aldehyde/ketone with α-H + dilute base → β-hydroxy carbonyl
Cannizzaro	aldehyde without α-H + conc. base → alcohol + carboxylate
Clemmensen / Wolff–Kishner	C=O → CH ₂ (Zn(Hg)/HCl · NH ₂ NH ₂ /KOH, glycol)
Rosenmund	RCOCl + H ₂ (Pd–BaSO ₄) → RCHO
Stephen	RCN + SnCl ₂ /HCl, then H ₃ O ⁺ → RCHO
HVZ	acid + X ₂ /red P → α-halo acid
Gabriel phthalimide	1° aliphatic amines only
Sandmeyer	ArN ₂ ⁺ + CuCl/CuBr/CuCN → ArCl/ArBr/ArCN
Carbylamine	1° amine + CHCl ₃ + KOH → isocyanide (foul smell)
Etard / Gattermann–Koch	toluene → benzaldehyde (CrO ₂ Cl ₂) · benzene + CO + HCl (AlCl ₃ /CuCl) → benzaldehyde

 REFERENCE Distinguishing tests (C13–C20) · page [MB 277](#)

Test	Positive for
Tollens' (silver mirror)	aldehydes (aliphatic and aromatic), formic acid, reducing sugars
Fehling's (red Cu_2O)	aliphatic aldehydes (not benzaldehyde)
Iodoform (I_2/NaOH)	CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$: ethanol, acetaldehyde, acetone, 2-propanol
Lucas (ZnCl_2/HCl)	turbidity: 3° at once · 2° in ~5 min · 1° none at room temperature
Neutral FeCl_3	violet colour with phenols
NaHCO_3	effervescence with carboxylic acids (not phenol, except picric acid)
Hinsberg reagent	1° : sulphonamide soluble in alkali · 2° : insoluble · 3° : no reaction
Lassaigne's test	N: Prussian blue · S: violet with nitroprusside · halogen: AgNO_3 precipitate

PART V · MASTER FORMULA HANDBOOK

Mathematics Formula Sheets

Σ FORMULA M01-M02 · Sets, functions, complex numbers & quadratics · pages [MB 287](#), [MB 292](#)

Result	Formula
Counting	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ · relations on an n -set: 2^{n^2} · reflexive: 2^{n^2-n} · symmetric: $2^{n(n+1)/2}$
Functions $A \rightarrow B$	total $ B ^{ A }$ · one-one mP_n ($n = A \leq m = B $) · onto from n to 2 elements: $2^n - 2$
Complex numbers	$ z_1 z_2 = z_1 z_2 $ · $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ · $z \bar{z} = z ^2$ · $ z_1 + z_2 \leq z_1 + z_2 $
Cube roots of unity	$1 + \omega + \omega^2 = 0$ · $\omega^3 = 1$ · $\omega = \frac{-1 + i\sqrt{3}}{2}$
Loci	$ z - z_1 = z - z_2 $: perpendicular bisector · $ z - z_0 = r$: circle
Quadratic	$\alpha + \beta = -\frac{b}{a}$ · $\alpha\beta = \frac{c}{a}$ · $D = b^2 - 4ac$ · $ \alpha - \beta = \frac{\sqrt{D}}{ a }$
Root location ($a > 0$)	both roots $> k$: $D \geq 0$, $f(k) > 0$, $-\frac{b}{2a} > k$ · k between the roots: $f(k) < 0$

Σ FORMULA M03 · Matrices & determinants · page [MB 298](#)

Result	Formula
Adjoint & inverse	$A^{-1} = \frac{\text{adj } A}{ A }$ · $A(\text{adj } A) = A I$
Determinant rules ($n \times n$)	$ kA = k^n A $ · $ \text{adj } A = A ^{n-1}$ · $ \text{adj}(\text{adj } A) = A ^{(n-1)^2}$ · $ A^{-1} = \frac{1}{ A }$
Products	$(AB)^{-1} = B^{-1}A^{-1}$ · $(AB)^T = B^T A^T$ · $\text{adj}(AB) = \text{adj } B \text{ adj } A$
Special matrices	symmetric $A^T = A$ · skew $A^T = -A$ (odd-order skew: $ A = 0$) · orthogonal $AA^T = I$
Linear systems $AX = B$	$ A \neq 0$: unique · $ A = 0$ and $(\text{adj } A)B \neq 0$: no solution · $ A = 0$ and $(\text{adj } A)B = 0$: infinitely many or none (check)
Area of a triangle	$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Σ FORMULA M04–M06 · P&C, binomial, sequences · pages MB 303, MB 308, MB 313

Result	Formula
Permutations & combinations	${}^nP_r = \frac{n!}{(n-r)!} \cdot {}^nC_r = \frac{n!}{r!(n-r)!} \cdot {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
Repetition	$\frac{n!}{p!q!r!} \cdot \text{circular } (n-1)! \cdot \text{necklace } \frac{(n-1)!}{2}$
Distribution	n identical into r groups: ${}^{n+r-1}C_{r-1}$ (each ≥ 1 : ${}^{n-1}C_{r-1}$)
Derangements	$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \cdot D_3 = 2, D_4 = 9, D_5 = 44$
Binomial general term	$T_{r+1} = {}^nC_r x^{n-r} y^r \cdot \text{middle term: } T_{n/2+1} \text{ (} n \text{ even)}$
Coefficient sums	$\sum {}^nC_r = 2^n \cdot \sum_{\text{odd}} = \sum_{\text{even}} = 2^{n-1}$
AP	$a_n = a + (n-1)d \cdot S_n = \frac{n}{2}[2a + (n-1)d]$
GP	$a_n = ar^{n-1} \cdot S_n = \frac{a(r^n - 1)}{r - 1} \cdot S_\infty = \frac{a}{1-r} \text{ (} r < 1 \text{)}$
Means	$AM \geq GM \geq HM \cdot \sum k^2 = \frac{n(n+1)(2n+1)}{6} \cdot \sum k^3 = \left[\frac{n(n+1)}{2} \right]^2$

Σ FORMULA M07 · Limits, continuity, differentiability & applications · page MB 318

Result	Formula
Standard limits	$\lim \frac{\sin x}{x} = 1 \cdot \lim \frac{e^x - 1}{x} = 1 \cdot \lim \frac{\ln(1+x)}{x} = 1 \cdot \lim \frac{a^x - 1}{x} = \ln a \cdot \lim \frac{x^n - a^n}{x - a} = na^{n-1}$
1^∞ form	$\lim f^g = e^{\lim g(f-1)}$
Expansions	$\sin x \approx x - \frac{x^3}{6} \cdot \cos x \approx 1 - \frac{x^2}{2} \cdot \tan x \approx x + \frac{x^3}{3} \cdot e^x \approx 1 + x + \frac{x^2}{2}$
Derivatives	$(\tan x)' = \sec^2 x \cdot (\sec x)' = \sec x \tan x \cdot (\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}} \cdot (\tan^{-1} x)' = \frac{1}{1+x^2} \cdot (a^x)' = a^x \ln a$
Rules	product, quotient, chain · parametric $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \cdot x^x$: take logs
Applications	rate of change · increasing where $f' > 0$ · maximum: $f' = 0, f'' < 0$ (Rolle/LMVT are not in the 2024+ text)

Σ FORMULA M08–M09 · Integration & differential equations · pages MB 324, MB 330

Result	Formula
Standard integrals	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} \cdot \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} \cdot \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $
More	$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln x + \sqrt{x^2 \pm a^2} \cdot \int e^x [f + f'] dx = e^x f \cdot \int \sec x dx = \ln \sec x + \tan x $
By parts	$\int uv dx = u \int v - \int (u' \int v) \text{ (ILATE)}$
Definite properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \cdot \text{even: } 2 \int_0^a \cdot \text{odd: } 0 \cdot \text{period } T: \int_0^{nT} = n \int_0^T$
Standard results	$\int_0^{\pi/2} \sin^n x dx \text{ (Wallis)} \cdot \int_0^{\pi/2} \ln \sin x dx = -\frac{\pi}{2} \ln 2 \cdot \int_0^{\pi/2} \frac{\sin^n}{\sin^n + \cos^n} = \frac{\pi}{4}$
Leibniz	$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h)h' - f(g)g'$
Areas	$y^2 = 4ax \text{ and } x^2 = 4by: \frac{16ab}{3} \cdot \text{ellipse } \pi ab \cdot \text{parabola } y^2 = 4ax \text{ with its latus rectum: } \frac{8a^2}{3}$
Linear DE	$y' + Py = Q: \text{IF} = e^{\int P dx} \cdot y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$

Σ FORMULA M10 · Coordinate geometry · page MB 335

Result	Formula			
Lines	distance $\frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}} \cdot \tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ · image: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$			
Circle	centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$ · chord $2\sqrt{r^2 - d^2}$			
Conic	e	Focus	Directrix	Latus rectum
$y^2 = 4ax$	1	$(a, 0)$	$x = -a$	$4a$
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)	$\sqrt{1 - \frac{b^2}{a^2}}$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$\frac{2b^2}{a}$
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\sqrt{1 + \frac{b^2}{a^2}}$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$\frac{2b^2}{a}$

Ellipse: $PS_1 + PS_2 = 2a$ · hyperbola: $|PS_1 - PS_2| = 2a$ · rectangular hyperbola $e = \sqrt{2}$.

Σ FORMULA M11-M12 · 3D geometry & vectors · pages MB 342, MB 348

Result	Formula
DCs	$l^2 + m^2 + n^2 = 1 \cdot \cos \theta = \frac{ a_1 a_2 + b_1 b_2 + c_1 c_2 }{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}}$
Shortest distance	skew $\frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$ · parallel $\frac{ (\vec{a}_2 - \vec{a}_1) \times \vec{b} }{ \vec{b} }$
Foot / image in a line	general point $Q(\lambda)$, then $\vec{PQ} \cdot \vec{b} = 0$ · image $2Q - P$
Products	$\vec{a} \cdot \vec{b} = a b \cos \theta$ · $ \vec{a} \times \vec{b} = a b \sin \theta$ · projection $\frac{\vec{a} \cdot \vec{b}}{ \vec{b} }$
Areas & volume	triangle $\frac{1}{2} \vec{AB} \times \vec{AC} $ · $[\vec{a} \vec{b} \vec{c}] = \text{volume}$ · coplanar iff $[\vec{a} \vec{b} \vec{c}] = 0$
Identities	$ \vec{a} \times \vec{b} ^2 + (\vec{a} \cdot \vec{b})^2 = a ^2 b ^2$ · $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Σ FORMULA M13-M14 · Statistics, probability & trigonometry · pages MB 353, MB 359

Result	Formula
Variance	$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$ · $y = ax + b$: $\sigma_y = a \sigma_x$ · first n naturals $\frac{n^2 - 1}{12}$
Grouped median / mode	$l + \frac{N/2 - C}{f} h$ · $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$
Probability	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ · $P(A B) = \frac{P(A \cap B)}{P(B)}$ · independent: $P(A \cap B) = P(A)P(B)$
Bayes	$P(E_k A) = \frac{P(E_k)P(A E_k)}{\sum P(E_i)P(A E_i)}$
Random variable	$E(X) = \sum xp$ · $\text{Var} = E(X^2) - [E(X)]^2$ · binomial: mean np , variance npq
Trig identities	$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$ · $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$ · $\sin 3A = 3 \sin A - 4 \sin^3 A$ · $\cos 3A = 4 \cos^3 A - 3 \cos A$
Range	$a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$
Inverse trig	$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ · $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ ($xy < 1$) · $\cos^{-1}(-x) = \pi - \cos^{-1} x$

PART V · MASTER FORMULA HANDBOOK

Constants, Units & Frequently Forgotten Formulas

VERIFIED FACT Physical constants

Exact values are fixed by the 2019 SI redefinition. The others are CODATA recommended values, rounded to the precision JEE uses [CODATA]. Use whatever values a question gives.

Constant	Value	JEE working value
Speed of light c	299 792 458 m/s (exact)	3×10^8 m/s
Planck constant h	$6.626\,070\,15 \times 10^{-34}$ J s (exact)	6.63×10^{-34} J s
Elementary charge e	$1.602\,176\,634 \times 10^{-19}$ C (exact)	1.6×10^{-19} C
Avogadro N_A	$6.022\,140\,76 \times 10^{23}$ mol $^{-1}$ (exact)	6.02×10^{23}
Boltzmann k_B	$1.380\,649 \times 10^{-23}$ J/K (exact)	1.38×10^{-23} J/K
Gas constant R	8.314 J mol $^{-1}$ K $^{-1}$	$8.314 \cdot 0.0821$ L atm · 0.0831 L bar · 1.987 cal
Faraday F	$96\,485$ C/mol	$96\,500$ C/mol
Gravitational G	6.674×10^{-11} N m 2 kg $^{-2}$	6.67×10^{-11}
ϵ_0	8.854×10^{-12} F/m	$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
μ_0	$1.256\,637 \times 10^{-6}$ N/A 2	$4\pi \times 10^{-7}$
Electron mass m_e	9.109×10^{-31} kg	9.1×10^{-31} kg
Proton mass m_p	1.6726×10^{-27} kg	1.67×10^{-27} kg
Atomic mass unit	1.6605×10^{-27} kg	931.5 MeV/ c^2
Rydberg R_∞	1.097×10^7 m $^{-1}$	13.6 eV (hydrogen ground state)
Stefan–Boltzmann σ	5.670×10^{-8} W m $^{-2}$ K $^{-4}$	5.67×10^{-8}
Wien's constant b	2.898×10^{-3} m K	2.9×10^{-3} m K

VERIFIED FACT Conversions & handy numbers

Quantity	Value
1 eV	1.6×10^{-19} J · $hc \approx 1240$ eV nm = 12400 eV Å
1 atm	$101\,325$ Pa = 1.01325 bar = 760 mm Hg
1 cal	4.184 J
$1 \text{ Å} \cdot 1 \text{ fm}$	10^{-10} m · 10^{-15} m
1 light year	9.46×10^{15} m
1 horsepower	746 W
$2.303RT/F$ at 298 K	0.0591 V
$\log 2 \cdot \log 3 \cdot \log 7 \cdot \ln 10$	$0.3010 \cdot 0.4771 \cdot 0.8451 \cdot 2.303$
$\sqrt{2} \cdot \sqrt{3} \cdot \sqrt{5} \cdot \pi^2$	$1.414 \cdot 1.732 \cdot 2.236 \cdot 9.87 (\approx g)$
$e \cdot e^{-1} \cdot \ln 2$	$2.718 \cdot 0.368 \cdot 0.693$

The frequently forgotten list RECOMMENDATION

These are the formulas students most often confuse or forget under time pressure, based on the traps collected in the modules. Revise this list in the last week and again on exam eve.

MISTAKE ALERT Physics

Often confused	Correct version
Lens vs mirror formula	lens $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ · mirror $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
Drop vs soap bubble	excess pressure $\frac{2T}{r}$ (drop, one surface) vs $\frac{4T}{r}$ (bubble, two surfaces)
Closed vs open pipe	closed: odd harmonics only, $f_1 = \frac{v}{4L}$ · open: all harmonics, $f_1 = \frac{v}{2L}$
g at height vs depth	height: $\left(1 - \frac{2h}{R}\right)$ · depth: $\left(1 - \frac{d}{R}\right)$
Axial vs equatorial dipole field	axial $\frac{2kp}{r^3}$ · equatorial $\frac{kp}{r^3}$ (and opposite to $\vec{p}$)
v_{rms} vs v_{mp}	$\sqrt{3RT/M}$ vs $\sqrt{2RT/M}$
Wire vs loop field	wire $\frac{\mu_0 I}{2\pi r}$ · loop centre $\frac{\mu_0 I}{2R}$
Moment of inertia of a rod	about the centre $\frac{1}{12}ML^2$ · about an end $\frac{1}{3}ML^2$
Photoelectric	K_{max} depends on frequency · current depends on intensity

MISTAKE ALERT Chemistry

Often confused	Correct version
Sign of work	chemistry (IUPAC): $\Delta U = q + w$, with w done on the system · physics: $\Delta Q = \Delta U + W$, with W done by the gas
Nernst sign	$E = E^\circ - \frac{0.0591}{n} \log Q$ (minus, and Q = products over reactants)
Molarity vs molality	per litre of solution vs per kg of solvent
$K_p = K_c(RT)^{\Delta n}$	Δn counts gaseous moles only
First-order half-life	independent of concentration · zero order: $t_{1/2} \propto [A]_0$
Carbocation vs carbanion order	opposite orders
NaBH_4 vs LiAlH_4	NaBH_4 does not reduce acids or esters
Fehling's vs Tollens'	benzaldehyde gives Tollens' but not Fehling's
Electron gain enthalpy	most negative: Cl, not F

MISTAKE ALERT Mathematics

Often confused	Correct version
Ellipse vs hyperbola	ellipse $b^2 = a^2(1 - e^2)$ · hyperbola $b^2 = a^2(e^2 - 1)$
$ kA $	$k^n A $, not $k A $
Distributing identical objects	${}^{n+r-1}C_{r-1}$ (with zeros allowed), not r^n
T_{r+1}	the term number is $r + 1$, so the 5th term uses $r = 4$
Principal range of $\cos^{-1}$	$[0, \pi]$, so $\cos^{-1}(-x) = \pi - \cos^{-1}x$
Variance under $y = ax + b$	$a^2\sigma^2$ (the constant b has no effect)
$\int_0^{2a} f$ with $f(2a - x) = f(x)$	$2 \int_0^a f$ (if $f(2a - x) = -f(x)$: zero)
IF of $y' - \frac{y}{x} = \dots$	$\frac{1}{x}$, not x

PART V · MASTER FORMULA HANDBOOK

Formula Revision Plans: 1-Day, 3-Day, 7-Day & Final 24 Hours

All four plans use the three-pass method (page 4): recall first, then check, then repeat what you missed. Reading a sheet without first trying to recall it feels productive, but it builds far less memory than testing yourself [LS-DUN], [LS-RK]. The plans are recommendations  **RECOMMENDATION**; adjust them to your weak areas from the Mistake Book.

The 7-Day Formula Cycle

Use this cycle once a month during preparation, and as the formula part of the final week.

Day	Session 1 (60 min)	Session 2 (45 min)	Re-test (15 min)	Done
1	Physics P01–P07	Maths M01–M03	yesterday's dots	☐
2	Physics P08–P12	Maths M04–M06	Day 1 dots	☐
3	Physics P13–P20	Maths M07–M09	Day 2 dots	☐
4	Chemistry C01–C08 (physical)	Maths M10	Day 1 + Day 3 dots	☐
5	Chemistry C09–C12 (inorganic) + NCERT tables	Maths M11–M14	Day 4 dots	☐
6	Chemistry C14–C20 (organic): reagent and named-reaction tables	Constants & frequently forgotten list	Day 2 + Day 5 dots	☐
7	Full self-test: 60 random formulas across all three subjects	Rewrite every missed formula 3 times	all remaining dots	☐

The 3-Day Formula Sprint

For the gap between two mocks, or when you have lost touch with the formulas after a break.

Day 1 · Physics

- ☐ Morning: sheets P01–P10 (recall, then check)
- ☐ Afternoon: sheets P11–P20
- ☐ Evening: frequently forgotten list (Physics); re-test the dots

Day 2 · Chemistry

- ☐ Morning: physical sheets C01–C08
- ☐ Afternoon: inorganic trends, then the reagent and named-reaction tables
- ☐ Evening: distinguishing tests; re-test the Day 1 and Day 2 dots

Day 3 · Maths

- ☐ Morning: M01–M07
- ☐ Afternoon: M08–M14, conic table from memory
- ☐ Evening: constants; the full dot list from all three days

The 1-Day Formula Blitz

For the day before a mock, or one revision day in the final fortnight. About 6 hours, with a 10-minute break every hour.

1

Hour 1 · Physics mechanics

P02–P06 sheets. Write the vertical-circle, banking, rolling and orbit results from memory.

2

Hour 2 · Physics rest

P07–P20. Focus on thermodynamics processes, optics sign conventions and the AC table.

3 Hour 3 · Physical chemistry

C01–C08. Write out every formula with its units.

4 Hour 4 · Inorganic & organic

Trend exceptions, the reagent table, named reactions and tests. Say each reaction aloud.

5 Hour 5 · Maths

M01–M14. Conic table, definite-integral properties, determinant rules, P&C distribution.

6 Hour 6 · Frequently forgotten + dots

All three trap tables, then everything you dotted today. Stop by evening and sleep normally.

The Final 24 Hours

☆ IMPORTANT The last 24 hours are for confidence and sleep, not new material

Research on sleep and memory consolidation shows that sleep after learning helps you retain it, and sleep deprivation impairs attention and working memory [LS-RB], [LS-AASM]. Do **not** study past your normal bedtime on the eve of the exam.

Time (for a morning shift)	Formula task
Eve, morning	7-Day cycle "Day 7" self-test, but only 30 formulas per subject
Eve, afternoon	Frequently forgotten list; the personal dot list from the last 7 days
Eve, evening (max 60 min)	Read the Last-Minute Revision Book sheets (page 36) once, calmly. No new problems.
Exam morning	Optional 10-minute glance at your personal "top 20 forgotten" card. Nothing else.
In the hall	Before starting, write any 3–5 formulas you tend to forget on the rough sheet (only after the exam officially starts).

REVISION Formula cycle log

Cycle	Dates	Formulas missed	Top 3 to fix	Done
1	_____	_____	_____	☐
2	_____	_____	_____	☐
3	_____	_____	_____	☐
4	_____	_____	_____	☐
5	_____	_____	_____	☐
6	_____	_____	_____	☐

Speed & Shortcut Manual

Legitimate speed methods (when to use, when not to, time saved, common mistake) and the answer-choice strategy.

• The JEE Main Speed & Shortcut Manual	26
• Calculation Fitness: Tables & Daily Drills	31

• Answer-Choice Strategy	33
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PART VI · SPEED & SHORTCUT MANUAL

The JEE Main Speed & Shortcut Manual

In JEE Main, "speed" is not fast writing. It is three separate skills:

1 · Selection

Recognising within about 20 seconds which questions to attempt now, later or never. This is the biggest time saver (Exam-Hall Protocol, page [MB 480](#)).

2 · Recognition

Seeing the question model ("this is a shortest-distance problem") so the method comes to mind at once. It comes from model drills in every chapter.

3 · Execution

Doing the algebra and arithmetic cleanly by hand, with no calculator. This manual and the drills in the next chapter train it.

The sixteen techniques below are **legitimate methods**: each one is either a mathematically valid shortcut or a way to eliminate options with certainty. Each gives when to use it, when **not** to use it, a worked example, the approximate time saved and the most common mistake. The time savings are this book's estimates  **RECOMMENDATION**, not measured data.

#	Technique	Main subjects
1	Dimensional analysis	Physics
2	Limiting cases	Physics, Maths
3	Special-value substitution	Maths
4	Back-substitution of options	Maths, Chemistry
5	Symmetry	Physics, Maths
6	One-significant-figure estimation	Physics, Chemistry
7	Ratio (proportionality) method	Physics, Chemistry
8	Conservation laws first	Physics
9	The standard-result bank	all
10	Graph instead of algebra	Maths, Physics
11	Equivalents and the n -factor	Chemistry
12	Log and pH shortcuts	Chemistry
13	Working backwards in organic chemistry	Chemistry
14	Elimination by impossibility	all
15	Casting out nines	all (arithmetic checks)
16	Cancel before you multiply	all

⚡ SHORTCUT 1 · Dimensional analysis**Use it when**

the options differ in the powers of physical quantities

Don't use it when

the options differ only in pure numbers ($2, \pi, \frac{1}{2}$) or in dimensionless groups

Example. The speed of a transverse wave on a string with tension T and linear mass density μ is (A) $\sqrt{T/\mu}$ (B) $\sqrt{T\mu}$ (C) T/μ (D) $\sqrt{\mu/T}$. $[T] = \text{MLT}^{-2}$ and $[\mu] = \text{ML}^{-1}$, so $[T/\mu] = \text{L}^2\text{T}^{-2}$ and only $\sqrt{T/\mu}$ is a speed.

Time saved: 1–2 minutes (no derivation).

Common mistake: treating angles, strains or e^x arguments as dimensional. Exponents and trig arguments must be dimensionless.

⚡ SHORTCUT 2 · Limiting cases**Use it when**

a formula depends on masses, angles or distances that can be pushed to 0, ∞ or equality

Don't use it when

the limit changes the physical situation (e.g. a block that stops sliding once μ gets large enough)

Example. For an Atwood machine, $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$. Check: $m_2 \rightarrow 0$ gives $a \rightarrow g$ (free fall) ✓; $m_1 = m_2$ gives $a = 0$ ✓. An option such as $\frac{(m_1 - m_2)g}{2m_1}$ passes the second test but fails the first ($m_2 \rightarrow 0$ gives $g/2$). Test two limits.

Time saved: 1–3 minutes on formula-choice MCQs.

Common mistake: using only one limit. Several options often pass the same one.

⚡ SHORTCUT 3 · Special-value substitution**Use it when**

the options are formulas in n, x or θ and the statement is an identity

Don't use it when

the options agree at every easy value, or the question asks for a count of solutions

Example. $\sum_{k=1}^n k \cdot k!$ equals (A) $(n+1)! - 1$ (B) $n! + 1$ (C) $(n+1)!$ (D) $n \cdot n!$. At $n = 1$ the sum is 1: (A) and (D) survive. At $n = 2$ the sum is $1 + 4 = 5$: (A) gives 5 and (D) gives 4. The answer is (A).

Time saved: 2–4 minutes compared with proving the result.

Common mistake: testing only $n = 1$ or $x = 0$, where several options coincide. Always test a second value.

⚡ SHORTCUT 4 · Back-substitution of options**Use it when**

the options are numbers and checking one is faster than solving

Don't use it when

numerical-value questions (there are no options), or options that contain parameters

Example. Solve $\sqrt{x+5} + \sqrt{x} = 5$: options 4, 9, 1, 16. At $x = 4$: $3 + 2 = 5$ ✓. No squaring is needed, so no extraneous roots can appear.

Time saved: 1–2 minutes.

Common mistake: accepting an option that "almost" works. The check must be exact.

⚡ SHORTCUT 5 · Symmetry**Use it when**

a circuit, charge distribution or integral is unchanged by a reflection, rotation or $x \rightarrow a + b - x$

Don't use it when

one element breaks the symmetry (a single different resistor, an asymmetric limit)

Example. $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$. Replacing x by $\frac{\pi}{2} - x$ swaps $\sin$ and $\cos$, so $2I = \int_0^{\pi/2} 1 dx$ and $I = \frac{\pi}{4}$.

Time saved: 3–5 minutes.

Common mistake: assuming a circuit is symmetric without checking every branch.

⚡ SHORTCUT 6 · One-significant-figure estimation**Use it when**

the options differ by a factor of 2 or more

Don't use it when

the options are within about 10% of each other, or you are subtracting nearly equal numbers

Example. Energy of a 500 nm photon: (A) 1.3×10^{-19} J (B) 4×10^{-19} J (C) 4×10^{-18} J (D) 4×10^{-20} J. Estimate: $\frac{(6.6 \times 10^{-34})(3 \times 10^8)}{5 \times 10^{-7}} \approx 4 \times 10^{-19}$ J. (In eV it is $1240/500 \approx 2.5$ eV. An option of 2.5×10^{-19} J would be a unit trap.)

Time saved: about 1 minute per calculation-heavy question.

Common mistake: estimating in mass-defect or small-difference problems, where the difference is the answer.

⚡ SHORTCUT 7 · Ratio (proportionality) method**Use it when**

the question changes one quantity and asks for the effect on another

Don't use it when

quantities combine additively ($R + r$, $u + at$), so no single power law applies

Example. A planet orbits at 4 times Earth's orbital radius. Since $T \propto r^{3/2}$, $T = 4^{3/2} = 8$ years. Chemistry: the time for equal volumes to effuse scales as $\sqrt{M}$, so O_2 takes $\sqrt{32/2} = 4$ times as long as H_2 .

Time saved: 1–2 minutes (no constants needed).

Common mistake: using the wrong power, such as $r^{2/3}$ instead of $r^{3/2}$.

⚡ SHORTCUT 8 · Conservation laws first**Use it when**

the question asks for a speed, height or final velocity and the forces are conservative (or collisions are involved)

Don't use it when

time, force or acceleration at an instant is asked, or friction does unknown work

Example. A bead slides from rest down a frictionless curved wire through a height h . Its speed at the bottom is $\sqrt{2gh}$, whatever the shape of the wire. Force analysis would need the curve's equation.

Time saved: 2–5 minutes.

Common mistake: conserving kinetic energy in an inelastic collision. Only momentum is conserved there.

⚡ SHORTCUT 9 · The standard-result bank**Use it when**

the situation matches a standard case exactly

Don't use it when

a condition differs (a projectile launched from a height, a non-uniform rod)

Example. Spin-only magnetic moments for $n = 1$ to 5 unpaired electrons are 1.73, 2.83, 3.87, 4.90 and 5.92 BM. For Fe^{3+} (high-spin d^5) you know 5.92 BM without computing $\sqrt{35}$. Other entries in the bank: $\int_0^{\pi/2} \ln \sin x \, dx = -\frac{\pi}{2} \ln 2$; the maximum range at 45° ; area between $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16ab}{3}$.

Time saved: 1–3 minutes each.

Common mistake: applying a standard result outside its conditions.

⚡ SHORTCUT 10 · Graph instead of algebra**Use it when**

the question asks for the number of solutions or the sign of a root

Don't use it when

the exact value of a root is required

Example. Number of real solutions of $e^x = x^2$. Sketch both curves. For $x < 0$ they cross once. For $x > 0$, $e^x - x^2$ is always positive (its minimum there is $2 - 2\ln 2 > 0$). Answer: 1.

Time saved: 2–4 minutes.

Common mistake: careless sketching near tangency or asymptotes. Check one or two exact points.

⚡ SHORTCUT 11 · Equivalents and the n-factor**Use it when**

titration, redox stoichiometry or electrolysis questions

Don't use it when

disproportionation or an unclear product, where the n -factor is ambiguous. Balance the equation instead

Example. Volume of 0.1 M KMnO_4 (acidic, $n = 5$) needed for 50 mL of 0.1 M FeSO_4 ($n = 1$): milliequivalents of $\text{Fe}^{2+} = 50 \times 0.1 \times 1 = 5$, so $V = \frac{5}{0.1 \times 5} = 10$ mL.

Time saved: 1–2 minutes (no balanced equation needed).

Common mistake: using $n = 5$ for KMnO_4 in neutral or weakly basic medium, where it is 3.

⚡ SHORTCUT 12 · Log and pH shortcuts**Use it when**

weak acids and bases with small α , and buffers

Don't use it when

very dilute solutions (below about 10^{-6} M), where water's own ions matter

Example. 0.01 M acetic acid with $\text{p}K_a = 4.74$: $\text{pH} = \frac{1}{2}(\text{p}K_a - \log C) = \frac{1}{2}(4.74 + 2) = 3.37$. A buffer with $[\text{salt}] = 10[\text{acid}]$ has $\text{pH} = \text{p}K_a + 1$.

Time saved: about 1 minute.

Common mistake: giving pH 8 for 10^{-8} M HCl. An acid solution can't be basic; the answer is just below 7.

⚡ SHORTCUT 13 · Working backwards in organic chemistry**Use it when**

the products are given and the starting compound is asked (ozonolysis, hydrolysis, degradation)

Don't use it when

carbocation rearrangements are possible. Then the forward mechanism is needed

Example. An alkene gives acetone and acetaldehyde on ozonolysis. Join the two carbonyl carbons with a double bond: $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$ (2-methylbut-2-ene).

Time saved: 1–3 minutes.

Common mistake: miscounting carbons. Each $\text{C}=\text{O}$ carbon came from one alkene carbon.

⚡ SHORTCUT 14 · Elimination by impossibility**Use it when**

always, as a first filter: probability > 1 , efficiency > 1 , negative speed or count, wrong sign of a quantity

Don't use it when

you are tempted to eliminate an option that merely *looks* unusual (a negative focal length is valid)

Example. Carnot efficiency between 500 K and 300 K: (A) 0.6 (B) 1.67 (C) 0.4 (D) 0.8. (B) is impossible, and $1 - \frac{300}{500} = 0.4$.

Time saved: varies; turns some four-way guesses into two-way decisions (Answer-Choice Strategy, page 33).

Common mistake: eliminating on "looks" instead of on a physical or mathematical impossibility.

⚡ SHORTCUT 15 · Casting out nines**Use it when**

checking a long multiplication or addition before you commit an NV answer

Don't use it when

as proof of correctness: it cannot detect swapped digits

Example. Is $347 \times 26 = 9022$? Digit sums reduce to $3 + 4 + 7 = 14 \rightarrow 5$ and $2 + 6 = 8$; $5 \times 8 = 40 \rightarrow 4$. $9 + 0 + 2 + 2 = 13 \rightarrow 4 \checkmark$. A mismatch would prove an error.

Time saved: prevents lost marks rather than saving time: about 15 seconds per check.

Common mistake: treating a match as proof. 9202 gives the same check digit.

⚡ SHORTCUT 16 · Cancel before you multiply**Use it when**

always: every fraction with products above and below

Don't use it when

—

Example. $\frac{22}{7} \times \frac{49}{11} \times \frac{3}{7}$: cancel 22 with 11 (giving 2) and 49 with 7×7 (giving 1), leaving $2 \times 3 = 6$. No four-digit numbers are ever written.

Time saved: 30–60 seconds per calculation, with fewer arithmetic slips.

Common mistake: rounding intermediate values (e.g. $\pi \approx 3$) and then being "between two options". Keep exact values until the last step.

PART VI · SPEED & SHORTCUT MANUAL

Calculation Fitness: Tables & Daily Drills

JEE Main allows no calculator (page [MB 10](#)). Hand calculation is a trainable skill, like fitness: 15 minutes a day for eight weeks makes a visible difference to both speed and error rate  **RECOMMENDATION**.

Numbers worth knowing by heart

Squares 11–30

n	n^2	n	n^2
11	121	21	441
12	144	22	484
13	169	23	529
14	196	24	576
15	225	25	625
16	256	26	676
17	289	27	729
18	324	28	784
19	361	29	841
20	400	30	900

Cubes & powers

n	n^3	2^n
6	216	64
7	343	128
8	512	256
9	729	512
10	1000	1024
11	1331	2048
12	1728	4096
13	2197	8192
14	2744	16384
15	3375	32768

Reciprocals

$\frac{1}{n}$	value	$\frac{1}{n}$	value
$\frac{1}{7}$	0.1429	$\frac{1}{14}$	0.0714
$\frac{1}{8}$	0.125	$\frac{1}{15}$	0.0667
$\frac{1}{9}$	0.1111	$\frac{1}{16}$	0.0625
$\frac{1}{11}$	0.0909	$\frac{1}{17}$	0.0588
$\frac{1}{12}$	0.0833	$\frac{1}{18}$	0.0556
$\frac{1}{13}$	0.0769	$\frac{1}{19}$	0.0526

Roots

$$\sqrt{2} = 1.414 \cdot \sqrt{3} = 1.732 \cdot \sqrt{5} = 2.236 \cdot \sqrt{6} = 2.449 \cdot \sqrt{7} = 2.646 \cdot \sqrt{10} = 3.162$$

$$\cdot \sqrt[3]{2} = 1.26 \cdot \sqrt[3]{3} = 1.442$$

Logs & exponentials

$$\log 2 = 0.301 \cdot \log 3 = 0.477 \cdot \log 5 = 0.699 \cdot \log 7 = 0.845 \cdot \ln 2 = 0.693 \cdot \ln 10 = 2.303 \cdot e^2 = 7.39 \cdot e^{-1} = 0.368$$

Factorials & trig

$$6! = 720 \cdot 7! = 5040 \cdot 8! = 40320 \cdot 10! = 3628800 \cdot \sin 37^\circ \approx 0.6 \cdot \cos 37^\circ \approx 0.8 \text{ (the 3–4–5 triangle)}$$

Mental-arithmetic methods

SHORTCUT Squares near a base

$(a \pm b)^2 = a^2 \pm 2ab + b^2$: $48^2 = (50 - 2)^2 = 2500 - 200 + 4 = 2304$. Numbers ending in 5: $n5^2 = n(n+1)$ followed by 25, so $65^2 = 4225$.

SHORTCUT Products near 100

97×94 : the deficits are 3 and 6. $97 - 6 = 91$ and $3 \times 6 = 18$, so the product is 9118. This works for any pair near a power of 10.

SHORTCUT Percentages & ratios

$x\%$ of $y = y\%$ of x : 16% of $25 = 25\%$ of $16 = 4$. Successive changes of $a\%$ and $b\%$ combine to $a + b + \frac{ab}{100}$ percent.

SHORTCUT Division by estimation

To divide by an awkward number, scale both parts first: $\frac{3.6}{0.012} = \frac{3600}{12} = 300$. Write powers of ten separately: $\frac{6 \times 10^{-7}}{3 \times 10^{-4}} = 2 \times 10^{-3}$.

The 15-minute daily drill RECOMMENDATION

1 Minutes 0–4 · Tables

Recall one table from this page (squares, cubes, reciprocals, logs) from memory. Check it; redo the misses.

2 Minutes 4–10 · Timed set

12 mixed calculations of the kind that appear in solutions: fraction simplification, powers of ten, a square root, a log. Write every step.

3 Minutes 10–13 · Algebra

3 expansions or factorisations, such as $(a + b + c)^2$ or $x^3 - y^3$, or completing the square.

4 Minutes 13–15 · Log it

Record time and errors in the tracker below. Aim for fewer errors first, then speed.

TIP Rotate the content

Week 1–2: arithmetic only. Week 3–4: add logs, powers and roots. Week 5+: take the calculations from your own mock solutions (the arithmetic you actually got wrong). This makes the drill specific to you.

REVISION Drill log (4 weeks)

Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Errors this week
1	☐	☐	☐	☐	☐	☐	☐	_____
2	☐	☐	☐	☐	☐	☐	☐	_____
3	☐	☐	☐	☐	☐	☐	☐	_____
4	☐	☐	☐	☐	☐	☐	☐	_____

PART VI · SPEED & SHORTCUT MANUAL

Answer-Choice Strategy

The options in an MCQ are information. Using them well is a legitimate exam skill. Guessing blindly is not: with -1 for a wrong answer, it adds risk for almost no expected gain.

The mathematics of attempting

Under the verified $+4 / -1$ scheme (page [MB 10](#)), suppose k options remain and you have no further information, so each is equally likely. The expected marks from attempting are:

$$E = \frac{4}{k} - \frac{k-1}{k} = \frac{5-k}{k}$$

Options remaining	Expected marks if you attempt	Probability of -1	Book rule  RECOMMENDATION
4 (pure guess)	+0.25	75%	Never.
3 (one eliminated)	+0.67	67%	Skip, unless the remaining three are not equally likely and your reasoning favours one
2 (two eliminated with certainty)	+1.50	50%	Attempt.
1 (solved)	+4.00	only if you erred	Attempt, then verify quickly

WARNING Why the book is stricter than the arithmetic

The table assumes your eliminations are **certainly right**. Under time pressure they often aren't, and every wrong answer also worsens your incorrect-to-correct ratio, which is a tie-breaker (page [MB 10](#)). Hence the rule used throughout this book: **attempt only after eliminating at least two options through real reasoning.**

IMPORTANT Numerical-value questions: never guess

Section B answers are typed in, so there is nothing to eliminate and a guess is almost never right. Enter an NV answer only after solving it. A blank costs 0; a wrong answer costs 1.

Seven legitimate ways to use the options

1 • Impossibility filter

Remove options that are physically or mathematically impossible: probability > 1 , efficiency ≥ 1 , negative counts, wrong sign of ΔG for a spontaneous process (Technique 14).

2 • Dimensions & units

In Physics, remove every option with the wrong dimensions (Technique 1).

3 • Limiting / special cases

Test each surviving option at a special value or limit (Techniques 2 and 3).

4 • Back-substitution

If the options are numbers, substitute them into the condition (Technique 4).

5 • Estimation

If the options are far apart, compute to one significant figure (Technique 6).

6 • Structure of the options

If two options differ only in sign or by a factor of 2, the question is testing that detail. Solve for that detail specifically.

7 · Consistency across parts

In "Statement I / Statement II" and "match the columns" questions, decide the item you are surest of first. It often removes two or three options at once.

Match-the-column and statement questions

These are common in Chemistry and Physics. Don't evaluate every pair in order.

STRATEGY Method

1. Find the **one pairing you are most certain of** and eliminate every option that contradicts it.
2. Repeat with the next most certain pairing. Usually two pairings leave one option.
3. For "Statement I and II" questions, decide each statement separately (true/false) before reading the options. Only then check whether II explains I, if the options ask for it.

What not to do**MISTAKE ALERT Myths that cost marks**

- **"Option C is correct most often."** There is no reliable answer-position pattern to exploit in NTA papers, and this book's own question bank deliberately has balanced keys.
- **"The longest option is usually right."** No evidence. It is a habit from poorly written tests.
- **"If I've left many questions, I should guess a few."** No. Skipped questions cost 0. The only way to raise your score is to solve more, or to eliminate with certainty.
- **Changing an answer on a "feeling"** during the final minutes. Change an answer only if you find a specific error in your working.

Mark-for-review discipline**SHORTCUT Use the review status as a queue, not a verdict**

Use **Marked for Review** only for questions you intend to return to, with a note on your rough sheet of how far you got ("Q14: $\lambda = 2$ found, need d "). In the last 10 minutes, review only questions where you already know the next step (Exam-Hall Protocol, page [MB 480](#)).

NOTE How marked answers are evaluated

In recent NTA instructions, a question that is answered and also "marked for review" is evaluated, while one only marked (with no answer) is not. Confirm this in the instructions on your screen before the exam starts.

THIRD-PARTY ANALYSIS

PART XII

Last-Minute Revision Book

Compressed one-page sheets for Physics, Chemistry and Maths, reaction and shortcut sheets, and the final checklists.

• How to Use the Last-Minute Book	36
• Physics in One Page	37
• Physical & Inorganic Chemistry in One Page	38
• Organic Reactions in One Page	39

• Maths in One Page	40
• Shortcut Checklist	41
• Mistake Checklist	42
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PART XII · LAST-MINUTE REVISION BOOK

How to Use the Last-Minute Book

Seven one-page sheets for the last three days and the exam morning. They hold **only the highest-yield lines**: the formulas, facts and rules that decide the most marks and are easiest to get wrong. The full versions are in the Formula Handbook (page [4](#)).

Sheet	Page	When to read
Physics in one page	37	D-3, D-1
Physical & inorganic chemistry in one page	38	D-3, D-1
Organic reactions in one page	39	D-2, D-1
Maths in one page	40	D-2, D-1
Shortcut checklist	41	D-2
Mistake checklist	42	D-1, exam morning
Strategy checklist	43	D-1, exam morning

 IMPORTANT Read, don't study

On these days, read each sheet once, calmly, and cover-test only the lines you mark as shaky. If a line looks unfamiliar, **don't** open the full chapter. Note it on your "top 5 forgotten" card and move on.

PART XII · LAST-MINUTE REVISION BOOK

Physics in One Page

Mechanics

$$R = \frac{u^2 \sin 2\theta}{g} \cdot H = \frac{u^2 \sin^2 \theta}{2g} \cdot T = \frac{2u \sin \theta}{g}$$

$$\text{Banking: } \tan \theta = \frac{v^2}{rg} \cdot \text{vertical circle: } v_{\text{bottom}} \geq \sqrt{5gl}$$

$$\text{Elastic, } m_2 \text{ at rest: } v_1 = \frac{m_1 - m_2}{m_1 + m_2} u, v_2 = \frac{2m_1}{m_1 + m_2} u$$

$$I: \text{ring } MR^2 \cdot \text{disc } \frac{1}{2}MR^2 \cdot \text{solid sphere } \frac{2}{5}MR^2 \cdot \text{rod } \frac{1}{12}ML^2 / \frac{1}{3}ML^2$$

$$\text{Rolling: } a = \frac{g \sin \theta}{1 + k^2/R^2} \cdot L = I\omega \text{ conserved if } \tau = 0$$

$$v_o = \sqrt{GM/r} \cdot v_e = \sqrt{2gR} \cdot T^2 \propto r^3 \cdot E = -\frac{GMm}{2r}$$

$$g_h \approx g(1 - 2h/R) \cdot g_d = g(1 - d/R)$$

Matter, heat, thermodynamics

$$\text{Bernoulli } v_t = \frac{2r^2(\rho - \sigma)g}{9\eta} \cdot h = \frac{2T \cos \theta}{r\rho g} \cdot \text{drop } \frac{2T}{r}, \text{ bubble } \frac{4T}{r}$$

$$W_{iso} = nRT \ln \frac{V_2}{V_1} \cdot \text{adiabatic } PV^\gamma \text{ const, } W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$\eta = 1 - \frac{T_2}{T_1} \cdot v_{rms} = \sqrt{\frac{3RT}{M}} \cdot \gamma = 1 + \frac{2}{f}$$

$$\text{Stefan } \sigma T^4 \cdot \text{Wien } \lambda_m T = b \cdot \text{Newton's cooling is exponential}$$

Oscillations & waves

$$T = 2\pi\sqrt{m/k} \cdot 2\pi\sqrt{l/g} \cdot v = \sqrt{T/\mu} \cdot \text{closed pipe: odd}$$

$$\text{harmonics, } f_1 = \frac{4L}{\lambda}$$

$$\text{Doppler } f' = f \frac{v \pm v_o}{v \mp v_s} \cdot \text{beats } |f_1 - f_2|$$

Electricity & magnetism

$$\text{Dipole: axial } \frac{2kp}{r^3}, \text{ equatorial } \frac{kp}{r^3} \cdot \text{sheet } \frac{\sigma}{2\epsilon_0} \cdot \text{conductor } \frac{\sigma}{\epsilon_0}$$

$$C = \frac{\epsilon_0 A}{d - t + t/K} \cdot U = \frac{1}{2}CV^2 \cdot \text{sharing: common } V = \frac{\sum \epsilon_0 V}{\sum C}$$

$$R \rightarrow n^2 R \text{ when stretched } n \text{ times} \cdot P_{max} \text{ at } R = r \cdot \text{meter bridge}$$

$$\frac{X}{R} = \frac{l}{100 - l}$$

$$B: \text{wire } \frac{\mu_0 I}{2\pi r} \cdot \text{loop } \frac{\mu_0 I}{2R} \cdot \text{solenoid } \mu_0 n I \cdot \text{arc } \frac{\mu_0 I \theta}{4\pi R}$$

$$r = \frac{mv}{qB} \cdot T = \frac{2\pi m}{qB} \cdot \text{shunt } S = \frac{I_g G}{I - I_g}$$

$$\epsilon = Blv \cdot \tau_{LR} = L/R \cdot Z = \sqrt{R^2 + (X_L - X_C)^2} \cdot \omega_0 = \frac{1}{\sqrt{LC}} \cdot P =$$

$$V_{rms} I_{rms} \cos \phi$$

$$\frac{E_0}{B_0} = c \cdot \text{spectrum: } \gamma, X, \text{ UV, visible, IR, micro, radio}$$

Optics

$$\text{Lens } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \cdot \text{mirror } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \cdot P = \sum P_i$$

$$\text{Prism } \delta_{min}: n = \frac{\sin \frac{A+\delta}{2}}{\sin \frac{A}{2}} \cdot \text{thin prism } \delta = (n - 1)A$$

$$\text{YDSE } \beta = \frac{\lambda D}{d} \cdot \text{slit central width } \frac{2\lambda D}{a} \cdot \text{Malus } \cos^2 \theta \cdot \text{Brewster } \tan i_B = n$$

Modern physics & devices

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})} \cdot K_{max} = h\nu - \phi = eV_0 \cdot \lambda_e = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \cdot \text{lines } \frac{n(n-1)}{2} \cdot R = R_0 A^{1/3} \cdot T_{1/2} = \frac{0.693}{\lambda}$$

$$\text{NAND, NOR universal } \cdot \overline{AB} = \bar{A} + \bar{B} \cdot \text{full-wave output } 2f$$

PART XII · LAST-MINUTE REVISION BOOK

Physical & Inorganic Chemistry in One Page

Physical chemistry

$$\Delta H = \Delta U + \Delta n_g RT \cdot \Delta G = \Delta H - T\Delta S \cdot \Delta G^\circ = -RT \ln K = -nFE^\circ$$

$$K_p = K_c(RT)^{\Delta n_g} \text{ (gases only)} \cdot Q < K: \text{forward}$$

$$\text{Weak acid } [H^+] = \sqrt{K_a C} \cdot \text{buffer pH} = pK_a + \log \frac{\text{salt}}{\text{acid}}$$

$$K_{sp}: AB \ s^2 \cdot AB_2 \ 4s^3 \cdot A_2B_3 \ 108s^5 \cdot \text{common ion lowers } s$$

$$\Delta T_f = iK_f m \cdot \pi = iCRT \cdot i = 1 + (n-1)\alpha$$

$$\text{Nernst } E = E^\circ - \frac{0.0591}{n} \log Q \cdot \log K = \frac{nE^\circ}{0.0591} \cdot \Lambda_m = \frac{1000\kappa}{M}$$

$$\text{First order } t_{1/2} = \frac{0.693}{k} \cdot \text{zero order } t_{1/2} = \frac{[A]_0}{2k} \cdot 87.5\% = 3 \text{ half-lives}$$

$$\text{Arrhenius } \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\text{Nodes: radial } n-l-1, \text{ angular } l \cdot E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$\text{Bond order } \frac{1}{2}(N_b - N_a) \cdot \text{paramagnetic: } O_2, B_2, O_2^- \cdot O_2^+ \text{ bond order } 2.5$$

Inorganic chemistry

IE exceptions: Be > B, N > O, Mg > Al, P > S · most negative EGE: Cl

Isoelectronic: $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

Shapes: XeF_2 linear · XeF_4 square planar · SF_4 see-saw · ClF_3 T-shaped · XeF_6 distorted octahedral

Basicity: H_3PO_4 3 · H_3PO_3 2 · H_3PO_2 1 · H_3BO_3 Lewis acid

Solid $PCl_5 = [PCl_4]^+ [PCl_6]^-$ · B_2H_6 : 3c-2e bonds

$KMnO_4$: acidic $n = 5$, neutral $n = 3$ · $K_2Cr_2O_7$ acidic $n = 6$

Spin-only μ : 1.73 · 2.83 · 3.87 · 4.90 · 5.92 BM ($n = 1$ to 5)

CFSE (oct) = $(-0.4t + 0.6e)\Delta_o$ · $\Delta_t = \frac{4}{9}\Delta_o$

Series: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$

$[Ni(CO)_4]$ tetrahedral, dia · $[Ni(CN)_4]^{2-}$ square planar, dia ·

$[NiCl_4]^{2-}$ tetrahedral, para

$[MA_4B_2]$: 2 geometrical · $[MA_3B_3]$: fac/mer · cis- $[M(AA)_2B_2]$:

optically active

Colourless: d^0, d^{10} ($Sc^{3+}, Ti^{4+}, Zn^{2+}$) · $KMnO_4$ colour = charge transfer

Lanthanoids: +3 common · Ce^{4+} oxidant · Eu^{2+}, Yb^{2+} reductants

· $Zr \approx Hf$

Cu^+ disproportionates in water · Mn: max +7 in the 3d series

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Organic Reactions in One Page

Reagents

$\text{H}_2/\text{Lindlar} \rightarrow \text{cis-alkene}$ · $\text{Na/liq. NH}_3 \rightarrow \text{trans-alkene}$
 $\text{HBr} + \text{peroxide} \rightarrow \text{anti-Markovnikov (HBr only)}$
 cold dilute alkaline $\text{KMnO}_4 \rightarrow \text{syn-diol (Baeyer)}$ · $\text{O}_3/\text{Zn-H}_2\text{O} \rightarrow \text{carbonyls}$
 LiAlH_4 reduces acids, esters, amides, nitriles · NaBH_4 : aldehydes and ketones only
 PCC : $1^\circ \rightarrow \text{aldehyde}$ · $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$: $1^\circ \rightarrow \text{acid}$, $2^\circ \rightarrow \text{ketone}$
 DIBAL-H : ester/nitrile $\rightarrow$ aldehyde · SOCl_2 : $\text{ROH} \rightarrow \text{RCl}$
 alcoholic KOH : elimination · aqueous KOH : substitution
 Br_2/NaOH on amide: Hofmann (one C fewer) · NaNO_2/HCl , 0–5 $^\circ\text{C}$: diazonium
 Sn/HCl : $\text{ArNO}_2 \rightarrow \text{ArNH}_2$ · Zn(Hg)/HCl or $\text{NH}_2\text{NH}_2/\text{KOH}$: $\text{C=O} \rightarrow \text{CH}_2$

Mechanism rules

SN_2 : methyl $> 1^\circ > 2^\circ > 3^\circ$, inversion · SN_1 : $3^\circ > 2^\circ > 1^\circ$, racemisation
 Carbocation $3^\circ > 2^\circ > 1^\circ$ · carbanion the reverse · EWG raises acidity
 Aqueous basicity: $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{aniline}$

Named reactions

Aldol: $\alpha\text{-H} + \text{dilute base}$ · Cannizzaro: no $\alpha\text{-H} + \text{concentrated base}$
 Reimer–Tiemann: phenol + $\text{CHCl}_3/\text{NaOH} \rightarrow \text{salicylaldehyde}$ · Kolbe: phenoxide + $\text{CO}_2 \rightarrow \text{salicylic acid}$
 Sandmeyer: $\text{ArN}_2^+ + \text{CuX} \rightarrow \text{ArX}$ · coupling with phenol $\rightarrow$ orange azo dye
 Gabriel: 1° aliphatic amines only · HVZ: $\alpha\text{-halo acid}$ · Wurtz: $2\text{RX} + 2\text{Na} \rightarrow \text{R-R}$
 Friedel–Crafts: AlCl_3 (not on aniline or nitrobenzene)
 Rosenmund: $\text{RCOCl} \rightarrow \text{RCHO}$ · Stephen: $\text{RCN} \rightarrow \text{RCHO}$ · Etard / Gattermann–Koch $\rightarrow$ benzaldehyde
 Williamson: $\text{RONa} + \text{R}'\text{X}$ ($\text{R}'\text{X}$ primary)

Tests

Tollens': all aldehydes, HCOOH · Fehling's: aliphatic aldehydes only
 Iodoform: CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$ (ethanol, acetone, 2-propanol)
 Lucas: 3° immediate, $2^\circ \sim 5 \text{ min}$, 1° none · neutral FeCl_3 : violet with phenols
 NaHCO_3 : CO_2 with acids · Hinsberg: 1° soluble in alkali, 2° insoluble, 3° no reaction
 Carbylamine: 1° amines only · Lassaigne: N gives Prussian blue
 Biomolecules: sucrose non-reducing · denaturation keeps 1° structure · RNA has U instead of T

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Maths in One Page

Algebra

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a} \cdot |z_1 z_2| = |z_1| |z_2| \cdot 1 + \omega + \omega^2 = 0$$

$$|kA| = k^n |A| \cdot |\text{adj } A| = |A|^{n-1} \cdot (AB)^{-1} = B^{-1} A^{-1}$$

$$|A| = 0 \text{ and } (\text{adj } A)B = 0 \rightarrow \text{check for infinitely many solutions}$$

$${}^n C_r, \frac{n!}{p!q!}, \text{circular } (n-1)! \cdot \text{identical into } r \text{ groups: } {}^{n+r-1} C_{r-1}$$

$$T_{r+1} = {}^n C_r x^{n-r} y^r \cdot \sum {}^n C_r = 2^n$$

$$\text{AP } S_n = \frac{n}{2}[2a + (n-1)d] \cdot \text{GP } S_\infty = \frac{a}{1-r} \cdot \text{AM} \geq \text{GM}$$

$$\text{Relations on } n \text{ elements: } 2^{n^2} \cdot \text{reflexive } 2^{n^2-n}$$

Calculus

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \cdot 1^\infty; e^{\lim g(f-1)} \cdot 1 - \cos kx \approx \frac{k^2 x^2}{2}$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2} \cdot (\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$\int e^x (f + f') = e^x f \cdot \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$\text{King: } \int_a^b f(x) = \int_a^b f(a+b-x) \cdot \text{odd function on } [-a, a]: 0$$

$$\text{Area } y^2 = 4ax, x^2 = 4by: \frac{16ab}{3} \cdot y = x^2 \text{ and } y = mx: \frac{m^3}{6}$$

$$\text{Linear DE: IF } = e^{\int P} \cdot P = \frac{1}{x} \rightarrow x \cdot P = \tan x \rightarrow \sec x$$

Coordinate geometry

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \cdot \text{parallel lines: equalise the coefficients first}$$

$$\text{Circle: centre } (-g, -f), r = \sqrt{g^2 + f^2 - c} \cdot \text{chord } 2\sqrt{r^2 - d^2}$$

$$\text{Parabola } y^2 = 4ax: \text{focus } (a, 0), \text{latus rectum } 4a$$

$$\text{Ellipse: } b^2 = a^2(1 - e^2) \cdot \text{hyperbola: } b^2 = a^2(e^2 - 1) \cdot \text{latus rectum } \frac{2b^2}{a}$$

3D & vectors

$$l^2 + m^2 + n^2 = 1 \cdot \text{SD} = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\text{Foot: } Q(\lambda) \text{ with } \vec{PQ} \cdot \vec{b} = 0 \cdot \text{image } 2Q - P$$

$$\text{Projection } \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \cdot \text{triangle } \frac{1}{2} |\vec{AB} \times \vec{AC}| \cdot |a \times b|^2 + (a \cdot b)^2 = a^2 b^2$$

Statistics, probability, trigonometry

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \cdot y = ax + b: \sigma_y = |a| \sigma_x \cdot \text{naturals: } \frac{n^2 - 1}{12}$$

$$\text{Bayes: chosen path} \div \text{sum of paths} \cdot E(X) = \sum xp \cdot \text{Var} = E(X^2) - \mu^2$$

$$a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}] \cdot \cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \cdot \sin^{-1}(\sin x) = x \text{ only on } [-\frac{\pi}{2}, \frac{\pi}{2}]$$

PART XII · LAST-MINUTE REVISION BOOK

Shortcut Checklist

Before solving any MCQ the long way, run down this list (Speed Manual, page [26](#)).

- | | |
|--|--|
| <ul style="list-style-type: none"> ☐ Dimensions: do the options differ in dimensions? Eliminate. ☐ Limiting cases: push a mass, angle or distance to 0, ∞ or equality. Test two limits. ☐ Special values: $n = 1, 2$ or $x = 0, \frac{\pi}{4}$ in identity-type options. ☐ Back-substitution: are the options numbers that are quick to check? ☐ Symmetry: can $x \rightarrow a + b - x$, or a symmetric circuit, cut the work? ☐ Estimation: are the options a factor of 2 or more apart? One significant figure. ☐ Ratios: is only one quantity changing? Use the power law. ☐ Conservation first: is a speed or height asked?
Energy/momentum before forces. | <ul style="list-style-type: none"> ☐ Standard results: is this a textbook case (rolling, area, μ values)? ☐ Graph: "number of solutions"? Sketch both sides. ☐ n-factor: titration or redox? Equivalents, not equations. ☐ pH: buffer? $pK_a + \log(\text{ratio})$. Very dilute? pH stays below 7 for an acid. ☐ Organic: products given? Work backwards. ☐ Impossibility: probability > 1? Efficiency ≥ 1? Wrong sign? ☐ Casting out nines on long NV arithmetic. ☐ Cancel before multiplying. Keep exact values until the end. |
|--|--|

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Mistake Checklist

The eight types: one-line prevention

- ☐ **CG · Concept:** am I sure which principle applies here?
- ☐ **FR · Recall:** does the formula's **condition** hold?
- ☐ **CA · Calculation:** every step written; verified by substitution or estimate?
- ☐ **MR · Misreading:** asked quantity and NOT/EXCEPT underlined?
- ☐ **TR · Transcription:** screen = rough sheet (letter, value, sign)?
- ☐ **TS · Time:** past 3 minutes? Leave it.
- ☐ **GU · Guessing:** two options eliminated? If not, don't attempt.
- ☐ **PR · Pressure:** stuck or panicking? 60-second reset.

The most costly slips

- ☐ $\text{cm} \leftrightarrow \text{m}$, $\text{kJ} \leftrightarrow \text{J}$, $\text{mL} \leftrightarrow \text{L}$, $^{\circ}\text{C} \leftrightarrow \text{K}$
- ☐ Lens vs mirror formula; drop vs bubble
- ☐ Closed pipe = odd harmonics only
- ☐ Δn_g counts gases only; n -factor of KMnO_4 depends on the medium
- ☐ Principal value of $\cos^{-1}$ lies in $[0, \pi]$
- ☐ The +1 when counting terms; the $\frac{1}{2}$ in triangle areas
- ☐ T_{r+1} : the 5th term uses $r = 4$
- ☐ d vs d^2 and radius vs diameter: answer what is asked
- ☐ "Incorrect statement" questions: pick the false one

My own repeats (from the Mistake Book):

PART XII · LAST-MINUTE REVISION BOOK

Strategy Checklist

Before the exam

- ☐ 8 hours of sleep on each of the last three nights
- ☐ Exam-day wake-up time used for all final mocks
- ☐ Admit card, photo ID and photos packed (admit card rules checked)
- ☐ Route and travel time checked; leave with a margin
- ☐ Familiar breakfast planned; water
- ☐ Command Sheet read (page [MB 483](#))

In the first 5 minutes

- ☐ Three slow breaths
- ☐ Rough sheet: skip columns and checkpoint times
- ☐ Memory dump of the 3–5 forgotten formulas

During the paper

- ☐ Subject order:
- ☐ Round 1: method visible in 20 s, ≤ 2 min, else **S**
- ☐ Round 2: 3 minutes maximum per question
- ☐ Checkpoints: 0:45 · 1:45 · 2:40
- ☐ MCQ only after eliminating two options; NV only if solved
- ☐ A hard section? Normalisation: it's hard for everyone. Next easy question.

The last 20 minutes

- ☐ Only questions with partial progress
- ☐ Every NV entry checked
- ☐ No answer changes without a specific error
- ☐ All responses saved